

NATIONAL SCHOOL OF STATISTICAL
SCIENCES
EST011 - INTRODUCTION TO
STATISTICS ASSIGNMENT
Group 1: activity A

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September 2, 2026

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Rio de Janeiro
09/01/2026

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1 What did we do?

We collected data using a methodology aimed at a fully automated collection process, so that we could define a way to quantify bias when inferring a tendency in the data collection process. To do this, we created two questions on the same topic — the university — and, on the same subject, about restructuring reforms at the university (in this case, our target audience being ENCE-IBGE students, the university in question specifically being ENCE-IBGE). The two questions are two versions of the same inquiry, one version (A): Neutral, and (B): Biased. To automate the data collection process, we aimed for an online means of answering the questions, using Forms. As a way to route the different versions of the question, we used website hosting, and HTML was the language used to build the site.

1.1 Where?

We hosted, on a website (Netlify), the redirection to two Forms links containing distinct versions of the question, which was shared in an ENCE-IBGE WhatsApp group containing all current and former students.

1.2 When?

The survey was conducted within the period from 08/24/2026 to 08/26/2026.

1.3 With whom or with what:

The target population was ENCE-IBGE students.

2 Formulas used:

2.1 Proportion of responses for A and B

$$\hat{p}_A = \frac{n_A^{yes}}{n_A} \quad \hat{p}_B = \frac{n_B^{yes}}{n_B} \quad (1)$$

2.2 Difference between the responses for A and B

$$d = \hat{p}_A - \hat{p}_B \quad (2)$$

3 The question and the result:

3.1 Version A (neutral):

Do you think the university needs to carry out reforms to its facilities?

3.2 Version B (biased):

Do you think the university should carry out reforms to its facilities in order to better meet your needs?

SAMPLE RESULTS		
Metric / Category	Version A	Version B
Answered yes	19	6
Total approached	25	26
Proportion of yes	0.76	0.80
COMPARATIVE ANALYSIS		
Difference A&B	=	-0.04
Refusals	=	26

3.3 Relevant Questions:

3.3.1 How did we randomize the version:

Refusals: 331. Randomization method: A website hosted using HTML that redirected to the Forms links at random. B: question B sounds biased because it mentions individual needs in relation to reforms at ENCE-IBGE, whereas in version A, individual need is not made explicit.

3.3.2 What we changed in the wording of version B:

We mentioned the individual needs of the student in relation to the reforms, instead of only generalizing the need for reforms.

4 Our phrase of up to 8 words.

Mentioning needs generates more certainty.

5 Did we use AI? For what?

1. To understand a concept from class, ask for an example, or clarify a term from the assignment.
2. To review the wording of a sentence that we wrote.
3. To check a calculation that we had already done by hand.
4. To discuss with the AI whether a definition we wrote was ambiguous, after having written it.
5. To ask for help structuring the code for this document in LaTeX.

6 What surprised us?

1. The willingness of students to answer the questionnaires was higher than expected.

7 An open question?

1. What improvements are desired by the participants?
2. Which sampling methodology would be most appropriate for the survey, especially considering the privacy of the users who provided the data?

8 Important Attachments: (x) Survey link (x) Code (X) Charts

8.1 Link to the survey:

Access the survey form through the: [Project Website](#).

8.1.1 HTML code for the form redirection:

```
1 <!DOCTYPE html>
2 <html>
3 <head>
4     <meta charset="utf-8"/>
5     <title>Redirecting</title>
6     <meta id="encurtador" http-equiv="refresh" content="5;
    url=/" />
7 </head>
8 <body>
9     <h1>Introduction to Statistics Assignment</h1>
10    <h2>Redirecting to FORMS</h2>
11    <script>
12        var links = [
13            Array('https://docs.google.com/forms/d/e/1
FAIpQLSd2wiAGmTh1TdIA53nGkhIkKpfWZR4fvpKMl2QsqDH_Ht2_ZA
/viewform?usp=dialog', 'https://docs.google.com/forms/d/
e/1FAIpQLSdAjuKGauA-
w98E1IFzQi_46LdCfZUriVqT3byf4R21nltXSv/viewform?usp=
dialog')
14        document
15            .getElementById("encurtador")
16            .setAttribute("content", "5; URL=" + links[Math.
    floor(Math.random()*links.length)]);
17    </script>
18 </body>
19 </html>
```

Code 1: HTML Language

8.2 Charts:

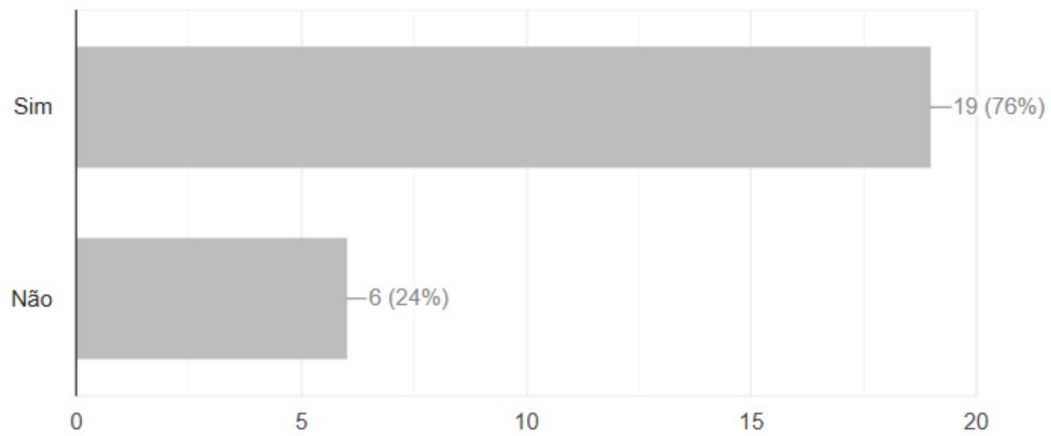


Figure 1: *Chart for question A*

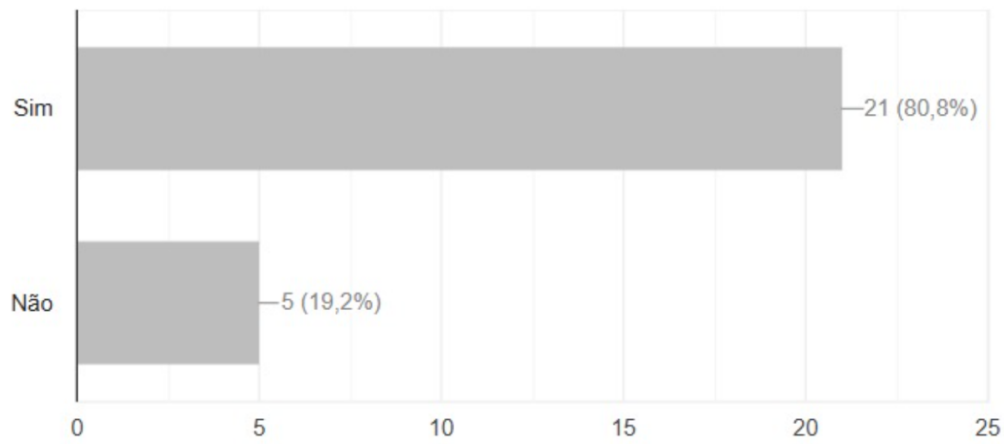


Figure 2: *Chart for question B*